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FIFTH SEMESTER B. TECH EXAMINATION, DECEMBER-2024

HELICOPTERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Attach Fig.1 (Power Requirements) with the answer book.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Identify *two* roles of mechanical transmission in single main rotor helicopter configuration..	L2	CO1	**02**
	b.	Describe autorotation flight of a helicopter.	L1	CO1	**02**
	c.	Define the parameter used to measure rotor efficiency in hover and its ideal value.	L1	CO3	**02**
	d.	Indicate *two* limitations of Momentum Theory	L2	CO4	**02**
	e.	Calculate the Disc Loading(DL) if ideal Power Loading(PL) of a given helicopter rotor operating at sea level is 70N/kW.	L3	CO3	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Name any *five* military applications of helicopters.	L1	CO1	**05**
		OR			
	b.	Name any *five* civilian applications of helicopters.	L1	CO1	**05**
3.	**a.**	Identify any five types of *disturbance* considered for static stability.	L2	CO2	**05**
		OR			
	b.	Explain the three dynamic stability states of a helicopter	L2	CO2	**05**
4.	**a.**	Derive a thrust expression during hover flight of the helicopter using momentum theory.	L3	CO3	**05**
		OR			
	b.	Illustrate the various velocity and force vectors acting on a blade element during vertical ascent using blade element theory.	L3	CO3	**05**
5.	**a.**	Explain estimation of induce inflow ratio ($\lambda i/\lambda_h$) for rotor angle of attack $\alpha=0^0$ in high speed forward flight.	L2	CO4	**05**
		OR			
	b.	Derive induced inflow ratio ($\lambda i/\lambda h$) against forward speed ratio ($\mu/\lambda h$) through a graph for disk angles of attack α, varying from + to – values and comment on the results.	L3	CO4	**05**

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6.	**a.**	Determine the endurance of a helicopter flying at 200kmph, carrying 900kg fuel and the engine SFC is 0.3 kg/kW/hr. The power corresponding to the speed is 750kW.	L3	CO5	**05**
		OR			
	b.	Predict the Range of the above helicopter flying at 200kmph.	L2	CO5	**05**
7.	**a.**	List five design features in tail rotor design.	L2	CO6	**05**
		OR			
	b.	Design the length of a moment arm for the tail rotor which has a thrust force of 400N. The main rotor power is 150kW and its RPM is 360.	L3	CO6	**05**
		PART-C			
				6 X 10=60	
8.	**a.**	Illustrate through a block diagram a single engine powered helicopter transmission system from engine to main rotor & tail rotor incorporating the following: **i.** Engine **ii.** Main Gear Box **iii.** Tail Gear Box **iv.** Intermediate Gear Box **v.** Tail Drive Shaft	L2	CO1	**10**
		OR			
	b.	Illustrate system of control axes and pitch, roll and yaw angles and the corresponding planes.	L2	CO1	**10**
9.	**a.**	Explain stages of autorotation In descent.	L2	CO2	10
		OR			
	b.	Explain safe height-velocity combinations during autorotation of a single engine and multi engine helicopter.	L2	CO2	**10**
10.	**a.**	Obtain expression for thrust over a rotor disk of radius R in hover using momentum theory.	L3	CO3	**10**
		OR			
	b.	Estimate the following power requirements at hover from Fig.1 **i.** Induced power **ii.** Profile power **iii.** Parasite power **iv.** Total power (check) **v.** Maximum Endurance power.	L3	CO3	**10**

11.	**a.**	Derive an expression for rotor thrust at high forward speed.	L3	CO4	**10**
		OR			
	b.	Calculate the thrust coefficient CT of a helicopter rotor blade having zero twist with solidity σ =0.106 and lift curve slope a=5.7, with blade pitch =10°.	L3	CO4	**10**
12.	**a.**	Derive an expression for disk angle of attack during forward flight of a Helicopter.	L3	CO5	**10**
		OR			
	b.	Explain estimation of total power during forward flight of a Helicopter.	L2	CO5	**10**
13.	**a.**	Discuss the factors leading to the occurrence of the following accidents in helicopters: **i.** Ground Resonance. **ii.** Dynamic rollover	L2	CO6	**10**
		OR			
	b.	Solve the pitch angle for a rotor blade with zero twist having the following data: No of blades=4; Rotor radius=6m; Chord=0.5m; lift curve slopes a=5.7; Rotor RPM=300; Mass of helicopter= 4000kg.	L3	CO6	**10**

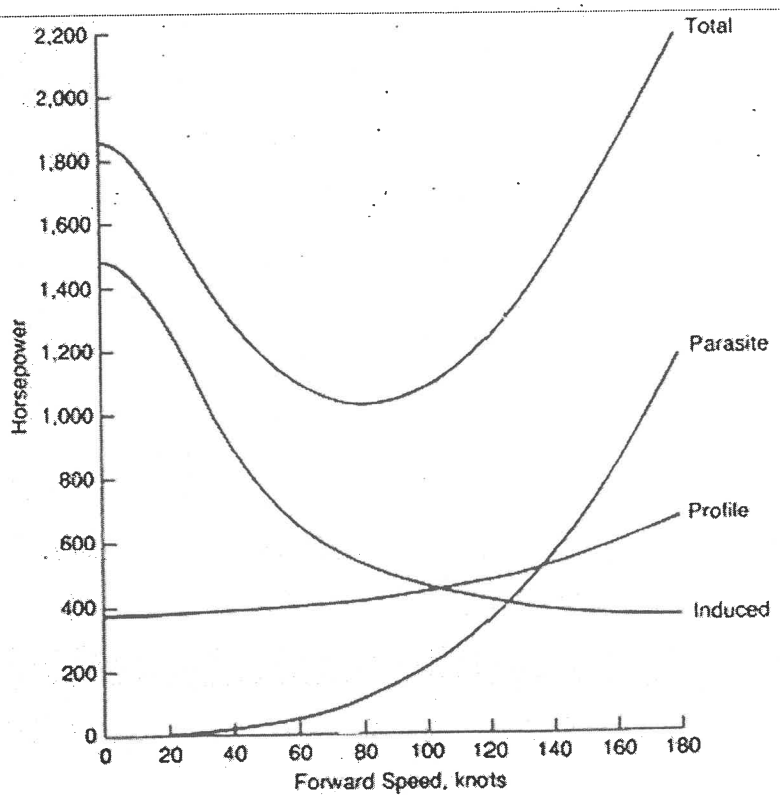

Fig. 1